

Abishek Singanur Aswan Kumar

M.S. Industrial Engineering — Operations Research, Optimization & Data-Driven Decision Systems
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RESEARCH PROFILE

M.S. Industrial Engineering researcher combining operations research, public-data engineering, research software, and AI-assisted information discovery. Experience curating large longitudinal datasets, validating source schemas, documenting provenance and derived measures, and building reproducible analytical systems, with growing research interest in geospatial metadata and public-data discovery.

RESEARCH INTERESTS

Public-data curation and research infrastructure · metadata, schema quality, provenance, and reproducibility · data discovery, indexing, and AI-assisted search · geospatial/network analytics · transportation and health applications.

EDUCATION

University of Illinois Urbana-Champaign

Aug 2025 – May 2027

M.S. Industrial Engineering · GPA 3.60 / 4.00 · Urbana, Illinois

Relevant coursework: Integer Programming; Algorithms for Data Analytics; Optimization of Large Systems; Operations Research for Manufacturing Systems; Data Science for Manufacturing Quality Control.

College of Engineering Guindy, Anna University

Nov 2021 – Jul 2025

B.E. Mechanical Engineering · Minor in Cybersecurity · GPA 3.60 / 4.00 (8.53 / 10) · Chennai, India

Indian Institute of Technology Madras

Sep 2021 – Dec 2026

B.S. Programming and Data Science · Chennai, India

Relevant coursework: Programming in Python; Statistics for Data Science; Machine Learning; Business Data Management; Tools for Data Science.

ACADEMIC APPOINTMENT

Teaching Assistant — IE 360: Facilities Planning and Design

Fall 2026

RESEARCH PUBLICATION

Design and Evaluation of a Novel Dual-Mode Four-Jaw Lathe Chuck Using Analytical Methods and FEA-Based Simulations.

2026

Srikanth, S.; Shameer, A.; Mahadev, B.; Nellanickal, R.; Kumar, A. S. *Production Engineering*, Springer. DOI: 10.1007/s11740-025-01376-5.

- Co-developed a dual-mode four-jaw chuck combining self-centering and independent jaw adjustment through an internal bevel-gear and pinion mechanism; derived torque, contact, stress, and clamping-force relations.
- Validated structural behavior through ANSYS stress, deformation, and modal analyses, demonstrating mechanical feasibility for flexible workholding under representative turning loads.

SELECTED RESEARCH & TECHNICAL PROJECTS

Airline Operations Intelligence Platform

Graduate research project · public-data engineering, operations research, metadata/provenance, and AI-assisted analytics

- Curated a longitudinal corpus of ~59M U.S. DOT/BTS flight records across **101 months (2018–2026)**, enabling reproducible carrier, airport, route, aircraft, delay, recovery, and network analyses.
- Reconstructed aircraft rotations as temporal operational sequences to study delay propagation, schedule recovery, and cascading disruption; quantified Southwest–American differences across **5 of 6** operational metrics.
- Engineered an **8-stage** ingestion, validation, normalization, and feature-engineering pipeline; audited the April **2026** schema and retained **119** meaningful BTS fields from **120** incoming columns to preserve semantics, provenance, and reuse.

- Developed an **11-page** research interface for structured carrier/airport/route/aircraft exploration and an LLM-assisted query layer that converts natural-language questions into traceable analytical operations.
- Documented source-field interpretations, operational definitions, data-quality rules, schema assumptions, and derived KPI logic to support reproducible dataset discovery and downstream research.

Healthcare MINCO — Hospital Operations Decision-Intelligence Platform

Optimization, forecasting, simulation, and explainable AI

- Developed a hospital-flow simulation, regime-detection, and robust-optimization framework; regime-robust control improved unsafe-excess outcomes over the nominal policy in **4 of 5** evaluation scenarios (average improvement **0.333**).
- Benchmarked the robust controller against a reinforcement-learning baseline, achieving mean reward **-32.8** vs. **-35.7**.
- Built a leakage-controlled forecaster for blocked arrivals (**R² 0.839**; **MAE 1.915**) and a balanced regime classifier reaching **80.7%** accuracy.
- Identified a **20%** ICU-capacity reduction as the highest-impact adverse scenario, increasing unsafe excess by **8.1** and blocked arrivals by **11.2**; developed counterfactual explanations.

Supply-Chain Vulnerability Intelligence Platform (SCVIP)

Independent research · network analytics and mixed-integer optimization

- Constructed a multi-echelon network with **35 suppliers, 8 components, 2 plants, 2 distribution centers, and 81 relationships**; evaluated sourcing policies across **500** correlated-disruption Monte Carlo scenarios.
- Maintained a comparable mean service level under hazard-aware and disruption-blind sourcing (81.6% vs 81.1%) while **reducing worst-case tail risk by ~64% relative to probability-only visibility**.
- Found that probability-only disruption information could underperform disruption-blind sourcing, indicating that incomplete risk visibility can produce ineffective diversification decisions.
- Formulated a fragility-aware mixed-integer program in Gurobi for supplier selection and procurement allocation under cost and capacity constraints, supported by graph-based vulnerability rankings and an executive recommendation layer.

Adaptive Tool-Holding System for Chatter Control

Capstone research · experimental manufacturing and machine learning

- Experimentally evaluated an adaptive magnetorheological-fluid boring bar with MEMS sensing and feedback control across **660–1,200 RPM** and multiple depths of cut.
- Reduced vibration by **~50% (>16 g to ~8 g)** and surface waviness by **~67% (~1.5 μm to ~0.5 μm)**; modeled stable regions via stability-lobe theory and a single-DOF model and implemented a decision-tree chatter classifier evaluated on $W_a/W_q/W_t/W_z$ measures.

AURUM — AI-Native Quantitative Research Platform

Independent research · optimization, simulation, and computational finance

- Implemented and compared **5 constrained portfolio optimizers** (minimum-variance, mean-variance, CVaR, Black-Litterman, LP-CVaR) across an **8-asset universe** with position, cash, and asset-class constraints.
- Built a three-state Gaussian hidden Markov regime engine and a **10,000-path Monte Carlo digital twin** to estimate VaR/CVaR and support constraint-compliant defensive reallocation; integrated SEC EDGAR / FRED data and a self-correcting research-memory loop.

ADDITIONAL RESEARCH & TECHNICAL PROJECTS

- **Solver-Based Optimization for Classification** — Formulated L1-regularized hinge-loss classification as a linear program and implemented direct and Benders-decomposition formulations in Gurobi; achieved **87.5% accuracy** with convergence in **19 master iterations**.
- **Yield and Throughput Optimization in Casting Operations** — Developed furnace-capacity, material-balance, and production-time models that increased output from **6 to 16 castings per mould**, reduced mould-preparation time **~66%**, and lowered the sand-to-metal ratio from **15.48 to 5.81**.
- **Robust Semiconductor Supply-Chain Optimization (in progress)** — Formulating a stochastic MILP for supplier activation, procurement allocation, and safety-stock decisions across **~200 correlated regional-disruption and supplier-yield scenarios**.

- **Nanoscale Metrology Micrometer System** — Designed an Arduino-controlled measurement system using microstepping, worm-gear reduction, and fine-pitch lead-screw motion — **~2.5 nm theoretical displacement resolution**, without laser interferometry.
- **Cost and Resource Optimization for Ship Operations** — Integrated telemetry, weather, routing, fuel, and maintenance data into scenario-based operational models that identified **\$1.5M+** in potential annual savings.
- **Weld Quality Classification** — Compared KNN, LDA, and QDA models on normalized manufacturing data using cross-validation, confusion matrices, RMSE, and feature-sensitivity analysis.

INDUSTRY EXPERIENCE

VDart — Gen AI / Recruitment Analytics Intern

Summer 2026 – Present

Remote

- Developing the structured-data layer of a Recruitment Intelligence Platform that converts Gmail traffic into normalized, queryable records; a 7-day snapshot classified **99** emails into **64 candidate submissions**, **27 hotlists**, and **8 replies**.
- Designed configurable information-extraction and classification logic across **10+** technology domains and multiple email types, translating semi-structured text into consistent fields and analytical endpoints.
- Implemented and tested Python/FastAPI services with SQLite/MySQL persistence, validation rules, pytest API checks, and frontend integration for repeatable ingestion and reliable downstream use.
- Extending the platform into a broader AI-enabled recruitment system where curated records support retrieval, candidate/job matching, ranking, recruiter analytics, and human review—making the current layer reusable decision-support infrastructure rather than a standalone script.

Tractor and Farm Equipment Ltd. (TAFE) — Engineering Intern

Jun 2023 – Jul 2023

Chennai, India

- Analyzed production, supplier, inventory, and delivery records in SAP ERP and built casting-production models incorporating furnace capacity, yield, pouring windows, and cycle-time constraints.
- Recommended a pattern configuration that raised output from **6 to 16 components per mould (+167%)** and reduced mould-prep time **66% (68.75 → 23.4 h)**; reduced the sand-to-metal ratio **15.48 → 5.81** via mass-balance analysis.
- Maintained tooling and engineering records in Windchill PLM, connecting shop-floor production analysis with design documentation and manufacturing process constraints.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, MATLAB, pandas, NumPy, scikit-learn, Jupyter; public-data acquisition, cleaning, schema profiling, EDA, feature engineering, statistical analysis

Data Curation & Research Infrastructure: metadata/data-dictionary documentation, schema mapping, validation, provenance, controlled-vocabulary/keyword tagging, indexing preparation, ETL, reproducible pipelines

Operations Research & Computational Methods: LP/MILP, robust optimization, Gurobi, HiGHS, Benders decomposition, scenario analysis, Monte Carlo simulation, network analytics

Research Software & Data Systems: FastAPI, Flask, Streamlit, REST APIs, SQLAlchemy, PostgreSQL, SQLite, MySQL, Docker, Redis, Git/GitHub; API/integration testing

AI, Classification & Search: machine learning, forecasting, information extraction, LLM-assisted analytics, natural-language querying, rule-based tagging, search-oriented interfaces

Research Practice: literature review, scholarly/technical writing, research documentation, source/metric definition, reproducible analysis, figure/table preparation, stakeholder communication

Engineering Tools: SAP ERP, Windchill PLM, ANSYS, CATIA, AutoCAD, CAD/CAM, CNC/FANUC, SPC, Lean, Six Sigma; route/network mapping.

LEADERSHIP & AWARDS

- Chairperson, Alumni Relations, SME — led an 8-member team; organized a 120–150-attendee alumni event.
- General Secretary, IoT Club, Anna University; Logistics Committee Member, PULSE, UIUC. Awards: Winner — IPL Auction Analytics, CCI Best Elevator Pitch, Interzonal Tennis.